

$$R = V/I = (4.8)/(0.040) = 120 \, \Omega$$

21 Ans: C

For 1 strand:

$$R = \frac{\rho l}{A} = (1.72 \times 10^{-8}) \frac{1.0}{\pi \left(\frac{0.3 \times 10^{-3}}{2} \right)^2} = 0.2433 \, \Omega$$

For 7 strands in parallel:

$$\frac{1}{R_{ll}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R}$$

$$R_{ll} = \frac{R}{7} = \frac{0.2433}{7} = 0.03476 \, \Omega$$

$$V = IR_{ll} = (13)(0.03476) = 0.4519 \, V$$

